

Fine-Tuning for Guarani Mbya: AI for Endangered Languages

The Language That Said “Wait” - Fine-Tuning AI for Guarani Mbya Preservation

Project Focus	The Language That Said “Wait” - Fine-Tuning AI for Guarani Mbya Preservation
Document Purpose	Organized copy of the project description provided by the user.

Visual Overview

Prefer learning visually? This video presents the Guarani Mbya fine-tuning challenge, the real-world context, and key concepts in visual format. It covers the same content as the text below—choose whichever format helps you learn best.

The Real-World Context

It's March 2023 at the Gwyrá Pepo Indigenous High School, two hours from São Paulo. A classroom of teenagers opens their laptops to test something no one has built before: an AI writing assistant for Guarani Mbya, their ancestral language.

These students speak Guarani Mbya fluently with their families. They tell stories, joke, argue, dream—all in their language. But when it comes to writing? They switch to Portuguese. That's what they learned first in school. That's what autocorrect knows. That's what feels easier on keyboards and phones. This is how languages die in the digital age: not with a bang, but with a thousand small switches to the dominant language.

IBM Research Brazil and the University of São Paulo approached the Tenondé Porã Indigenous community with a proposal. What if we built an AI writing assistant for your language? Something like autocorrect, but trained on Guarani Mbya. The students could write in their language, and the AI would help complete sentences, suggest words, catch mistakes. The community agreed to a trial.

The Technical Challenge Nobody Talks About

This wasn't like fine-tuning a model for Spanish or French. Guarani Mbya is classified by UNESCO as a 'vulnerable language'—around 17,000 speakers, mostly restricted to home and family contexts. There's barely any digital text. No Wikipedia corpus to scrape. No millions of training examples. No pre-trained models that understand its grammar.

The researchers had to fine-tune pre-trained models on what they could gather: a Guarani Mbya dictionary compiled by linguists over decades, educational materials from the community, and translations of religious texts. They used techniques like LoRA and QLoRA to fine-tune efficiently with minimal data, training the model slowly to avoid memorization instead of actual learning.

And then there's the grammar problem. Guarani Mbya places verb tense information at the start of words, not the end. In English, we say 'he does'—the action comes first. In Guarani Mbya, it's more like 'he esdo' where the tense marker comes as a prefix. This makes word prediction exponentially harder because the AI has to 'look past' prefixes to understand root words and their meanings. Every autocomplete prediction requires understanding grammatical structure that mainstream language models were never designed to handle.

The 'Linguistic Assistant' app worked like autocorrect on smartphones—students started typing and the AI suggested completions based on what it learned. Teachers reported progress. Students wrote longer sentences in Guarani Mbya instead of defaulting to Portuguese. From a technical standpoint, the fine-tuned model was succeeding despite the minimal training data.

Then The Community Said: Stop

After four months of testing in the classroom, the Tenondé Porã community held internal discussions. They came back to IBM with a clear message: We're concerned about exposing our language and culture to the internet. We need to pause this project while we discuss it as a community.

They weren't rejecting the technology outright. They were asserting their right to deliberate collectively about whether digital tools align with their cultural values. They were asking the questions that matter: Who owns this AI once it's trained on our language? Who benefits from it? What happens to our knowledge once it's digitized and potentially accessible to tech companies? What if our language data gets extracted and used without our consent? Do we want our children learning to write from an algorithm, or from elders and teachers?

IBM researchers Dr. Claudio Pinhanez and his team could have pushed back. They could have cited their partnership agreement, pointed to measurable student progress, emphasized the urgency of language preservation. UNESCO reports that nearly half of the world's 7,000 languages are expected to go extinct by 2100. Every day without digital tools means more languages slip closer to disappearance.

Instead, they said: 'No language data was collected during the process. Thank you for the opportunity to collaborate with your students. We're suspending work on the Guarani Mbya writing assistant, accordingly.' And then they waited. The project remains paused as of 2025. The community is still discussing. The researchers are still waiting. The fine-tuned model sits unused.

'Nothing for us without us.' — Declaration of Los Pinos, UNESCO International Decade of Indigenous Languages

The Toxic Data Problem

While working on Guarani Mbya, the IBM team also developed tools for Nheengatu, another endangered Brazilian Indigenous language with around 20,000 speakers. Here they confronted an ethical nightmare that every AI practitioner eventually faces: What do you do when your only training data is poisoned?

For Nheengatu, the largest available text corpus was Bible translations. Translated by colonial Portuguese priests between 1600 and 1800, during the era when Portugal was systematically outlawing and suppressing Indigenous languages. The Nheengatu language itself emerged as a lingua franca during colonization—a way for Indigenous communities to communicate with Portuguese settlers and each other. Later, it was banned twice: when settlers consolidated power and again when Brazil became independent in 1822.

The research team classified this data as potentially 'toxic'—not because the Bible itself is toxic, but because these specific translations used archaic language that native speakers don't recognize, reflected the cultural violence and abuse of the colonial period, and represented the very power structure that tried to erase these languages. Fine-tune on colonial texts and you risk teaching the AI language patterns that perpetuate historical oppression.

So they experimented carefully. They fine-tuned models using only the colonial Bible data, then tested translations with native speakers. They discovered that multilingual models trained on many Indigenous languages simultaneously achieved impressive accuracy scores but were actually 'cheating'—memorizing Bible verses instead of learning Guarani Mbya grammar. Perfect BLEU scores on benchmarks, but useless real-world performance when students tried to write original sentences.

The solution wasn't clean. They used the Bible translations but acknowledged the contamination. They tested extensively with native speakers. They prioritized data quality over quantity. They trained models slowly to prevent rogue memorization. They documented which outputs showed traces of colonial language patterns. This is fine-tuning in the real world: messy, historically loaded, ethically fraught decisions that require constant human judgment.

Your Challenge

Fine-tuning isn't just about improving accuracy metrics on clean Kaggle datasets. It's about deciding whose knowledge the model learns, whose perspective it amplifies, and who gets to say 'wait.' The Guarani Mbya case reveals that fine-tuning is fundamentally a question of power.

The big question: Can you build an AI system where the people whose knowledge you're learning from have more power than you do?

Assignment: Challenge - Fine-Tune on Minimal Data

Recreate what IBM did for the Guarani Mbya students.

Build a working writing assistant for a low-resource language using fine-tuning techniques. Pick any language with limited digital resources—an endangered Indigenous language if you have access to linguistic materials, or simulate the constraints by artificially limiting your training data for a better-resourced language. The technical challenge is identical: how do you fine-tune effectively when you only have a few thousand examples instead of millions?

Your task starts with gathering minimal training data. Find or create a small dataset of 1,000-5,000 text examples—perhaps dictionary entries, educational materials, or translated documents. Then choose a small pre-trained model you can actually fine-tune locally, something like GPT-2 small or BLOOM-560m. Apply efficient fine-tuning techniques like LoRA or QLoRA that work with limited data and computational resources. Build a simple interface that demonstrates word completion or sentence prediction, similar to what the Guarani Mbya students used in their classroom.

The critical technical question is whether your model actually learns language patterns or just memorizes your training examples. IBM discovered that multilingual models were 'cheating' by storing Bible verses instead of learning grammar. You need to test for this. Create evaluation examples that don't appear in your training data. Check if the model can generalize to new contexts or if it only repeats what it's seen before. Measure quality versus memorization—this distinction matters enormously when you're building tools for real communities to preserve their languages.

Success means building a working fine-tuned model with a demo interface, testing it with realistic examples from native speakers or domain experts, and documenting everything in a technical report. What techniques helped with minimal data? Did your model memorize or generalize? What would you need to make this production-ready for actual community use? The evaluation criteria isn't just 'does it work?' but 'would you feel confident giving this to Indigenous teenagers to help preserve their language?'

Your research territory: Low-resource language fine-tuning, LoRA and QLoRA techniques, evaluation metrics that detect memorization versus learning, building interfaces for language learners.

Ethics & Responsibility

The Guarani Mbya case forces questions that don't have easy answers. Who owns a language? When you fine-tune a model on Guarani Mbya text, who owns the resulting AI? IBM? The Tenondé Porã community? Individual speakers? The teenagers who tested it? OpenAI or Meta who made the base model? What if commercial products later use your fine-tuned model without permission?

Consider the urgency versus autonomy dilemma. Nearly half of the world's languages are expected to go extinct by 2100. Every day without AI tools means more languages slip closer to disappearance. But rushing to 'save' languages without community consent recreates colonial patterns of extraction—exactly what suppressed these languages in the first place. How do you balance urgency with autonomy? Who decides whether speed or sovereignty matters more?

Think about what digital preservation really means. Putting a language online makes it accessible for future generations and enables tools like the writing assistant the students tested. But it also exposes the language to misuse, mockery, commercial exploitation, and loss of control. Is digitization always good for endangered languages? What if elders prefer oral transmission? What if the community values keeping certain knowledge private? Who gets to decide these trade-offs?

And here's the question that should keep you up at night: Can you build an AI system where the people whose knowledge you're learning from have more power than you do? When OpenAI fine-tunes GPT on your conversations, can you pause the project? When Google fine-tunes models on your photos, do you have a veto? When your employer fine-tunes models on your work, can you withdraw consent? Most of us can't. We click 'I Agree' and hope for the best.

The Guarani Mbya community could pause the project because IBM researchers built a partnership that respected their sovereignty. They didn't just extract data and leave. They stayed. They waited. They're still waiting. This is what responsible fine-tuning looks like—not just better accuracy metrics, but better power dynamics. Not just optimizing models, but respecting the people whose knowledge trains them.

Why This Matters

Fine-tuning is how foundation models become specialized for specific domains, languages, and tasks. Every enterprise AI deployment involves fine-tuning on proprietary data. Every domain-specific model like Med-PaLM or CodeLlama started with fine-tuning. Understanding fine-tuning means understanding how AI actually gets deployed in the real world.

But the Guarani Mbya case reveals that fine-tuning is never just technical. It's political. It's about whose knowledge gets encoded into AI systems, whose perspective gets amplified, and who benefits from the resulting models. When you fine-tune on minimal data, work with toxic historical texts, or build tools for marginalized communities, every technical decision carries ethical weight.

The challenge isn't just making fine-tuning work with limited data. It's building AI systems where the people most affected have the power to say 'wait'—and knowing that if they do, you'll listen. That's the kind of AI practitioner the world needs: someone who can optimize a model and also recognize when optimization isn't the point.

Getting Started

Start by understanding the domain and the community. If you're working with Indigenous languages, connect with linguists and language revitalization programs. If you're working with historical data, consult historians and affected communities. If you're working with medical or legal data, talk to practitioners who understand the domain's ethical complexity. Never assume you understand the context just from reading datasets.

For technical resources, explore IBM Research's publications on Guarani Mbya and Nheengatu fine-tuning for detailed methodologies. Study low-resource language techniques like LoRA, QLoRA, and parameter-efficient fine-tuning. Learn about evaluation metrics that distinguish memorization from learning. Investigate frameworks for Indigenous data sovereignty and community-centered AI design.

Pick your challenge level based on where you want to grow. Level 1 teaches technical fundamentals of fine-tuning with minimal data. Level 2 confronts the ethical complexity of working with poisoned training data. Level 3 reimagines power dynamics in AI deployment. All three require both technical skill and ethical reflection. Document everything in your decision log, especially the decisions that don't have clear right answers.